

Total No. of Questions : 8]

SEAT No. :

P6780

[Total No. of Pages : 2

[6181]-406

B.E. (Artificial Intelligence and Data Science)

QUANTUMARTIFICIALINTELLIGENCE

(2019 Pattern) (Semester - VII) (417523A) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Describe Shor's Algorithm with example. [6]
b) Explain Simon's Algorithm. [6]
c) Explain Factoring Integers, with example. [6]

OR

- Q2)** a) Describe Grover's Algorithm with example. [6]
b) Discuss analysis of Deutsch algorithm in details. [6]
c) Write a short note on: Phase Kick-Back. [6]

- Q3)** a) Explain in details quantum support vector machines. [6]
b) Explain Nearest Neighbour Search in detail. [6]
c) Explain in detail - Quantum Boosting. [6]

OR

- Q4)** a) Write a short note on: Variational Quantum Algorithms. [6]
b) Explain Quantum Neural Networks. [6]
c) Describe classification in quantum machine learning with examples. [6]

- Q5)** a) What are some practical challenges and limitations associated with error model encoding in quantum computing. [6]
b) Describe Quantum Key Distribution in short. [6]
c) Discuss the Classical Three-Bit Code. [5]

OR

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- Q6)** a) Explain in short Quantum Dense Coding. [6]
b) Discuss in detail Quantum cryptography. [6]
c) Write a short Note on Fault Tolerance. [5]

- Q7)** a) Write short note on Quantum Walk. [6]
b) Explain Quantum Neural Computation. [6]
c) Write a short note on Quantum Tree Search. [5]

OR

- Q8)** a) State the challenges and limitations are associated with quantum computing in data sciences. [6]
b) Explain Quantum Production System in detail. [6]
c) Describe the Heuristic Search with example. [5]

